LECTURE: 5 INTRODUCTION TO STL

Mondira Chakraborty

Lecturer, CSE Dept.

Notre Dame University Bangladesh

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1. What is STL?
2. Key components of STL
3. STL-Container
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What is STL?

- It is a collection of algorithms, data structures, and other components that can be used to simplify the development of C++ programs.

- The STL provides a range of containers, such as vectors, lists, and maps, as well as algorithms for searching, sorting and manipulating data.

- Many of the algorithms and data structures in the STL are implemented using optimized algorithms, which can result in faster execution times compared to custom code.

Key Components of STL

- **Containers:** The STL provides a range of containers, such as vector, list, map, set, and stack, which can be used to store and manipulate data.

- **Algorithms:** The STL provides a range of algorithms, such as sort, find, and binary_search, which can be used to manipulate data stored in containers.

Key Components of STL

- **Iterators:** Iterators are objects that provide a way to traverse the elements of a container.

- **Function Objects:** Function objects, also known as **functors**, are objects that can be used as function arguments to algorithms. They provide a way to pass a function to an algorithm, allowing you to customize its behavior.

Container

- Containers or container classes store objects and data
- We can classify STL containers into 4 types:

I. **Sequence Containers:** implement data structures that can be accessed in a sequential manner.
   - vector
   - list
   - deque
   - arrays
   - forward_list(Introduced in C++11)

Container

II. **Container Adaptors:** provide a different interface for sequential containers
   - queue
   - priority_queue
   - stack

III. **Associative Containers:** implement sorted data structures that can be quickly searched
   - set
   - multiset
   - map
   - multimap

Container

IV. **Unordered Associative Containers:** implement unordered data structures that can be quickly searched

- unordered_set (Introduced in C++11)
- unordered_multiset (Introduced in C++11)
- unordered_map (Introduced in C++11)
- unordered_multimap (Introduced in C++11)

Algorithms

- An algorithm is a series of instructions to solve a particular problem.
- In C++, we can use the Standard Template Library to implement some of the commonly used algorithms
- If one's goal is to succeed in competitive programming, knowing merely the ins and outs of STL containers will not let them fully utilize everything that STL has to offer.
- STL has an ocean of algorithms, for all < algorithm > library functions
- These STL components are simply known as the "algorithms library"

Algorithms

- They act on containers and provide means for various operations for the contents of the containers.
- Algorithm
  - Sorting
  - Searching
  - Important STL Algorithms
  - Useful Array algorithms
  - Partition Operations

Iterators

- An **iterator** is an object (like a pointer) that points to an element inside the container.
- They can be visualized as something similar to a pointer pointing to some location and we can access the content at that particular location using them.
- Iterators are used to point at the memory addresses of STL containers
- They are primarily used in sequences of numbers, characters etc.
- They reduce the complexity and execution time of the program.

Functors

- Consider a function that takes only one argument. However, while calling this function we have a lot more information that we would like to pass to this function, but we cannot as it accepts only one parameter. What can be done?
- One obvious answer might be global variables.
- However, good coding practices do not advocate the use of global variables and say they must be used only when there is no other alternative.
- **Functors** are objects that can be treated as though they are a function or function pointer.

References:

1. https://www.geeksforgeeks.org/cpp-stl-cheat-sheet/?ref=lbp
2. https://usaco.guide/bronze/intro-ds?lang=cpp
3. https://www.programiz.com/cpp-programming/standard-template-library
4. https://www.geeksforgeeks.org/the-c-standard-template-library-stl/
5. https://www.geeksforgeeks.org/c-magicians-stl-algorithms/4
6. https://www.geeksforgeeks.org/functors-in-cpp/